# Power Budget: Force Sensing Board

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|---|---|
| **Team Number:** | 103 |
| **Project Name:** | **GripRx** |
| **Team Member Names:** | **Duotao Gao, Gabriel Toneser Facchin, Zice Sun** |
| **Version:** | **2 (Zice Sun, 2026-10-26: MCP6004 removed per instructor feedback)** |

***A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements***

| All Major Components | Component Name | Part Number | Supply Voltage Range | # | Absolute Maximum Current (mA) | Total Current (mA) | Unit |
|---|---|---|---|---|---|---|---|
| | PIC18F57Q43 Curiosity Nano (MCU board) | DM164150 | 1.8 - 5.5 V | 1 | 350.0 | 350.0 | mA |
| | Instrumentation amplifier | INA125P | 2.7 - 36 V | 1 | 12.5 | 12.5 | mA |
| | Load cell (bridge excitation) | 3135_0 (Phidgets) | 5 V max | 1 | 5.0 | 5.0 | mA |
| | | | | | | 0.0 | mA |

***B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design. Add additional power rails or change the power rail voltages if needed.***

| *+12V Power Rail (not used)* | Component Name | Part Number | Supply Voltage Range | # | Absolute Maximum Current (mA) | Total Current (mA) | Unit |
|---|---|---|---|---|---|---|---|
| | None on this board | — | — | 0 | 0 | 0 | mA |
| | | | | | | 0 | mA |
| | | | | | | 0 | mA |
| | | | | | | 0 | mA |
| | | | | | ***Subtotal*** | 0 | mA |
| | | | | | ***Safety Margin*** | 25% | |
| | | | | | ***Total Current Required on +12V Rail*** | 0 | mA |
| ***c1. Regulator or Source Choice*** | Not used (no +12 V loads) | — | — | 0 | 0 | 0 | mA |
| | | | | | ***Total Remaining Current Available on +12V Rail*** | 0 | mA |

| *+5V Power Rail* | Component Name | Part Number | Supply Voltage Range | # | Absolute Maximum Current (mA) | Total Current (mA) | Unit |
|---|---|---|---|---|---|---|---|
| | PIC18F57Q43 Curiosity Nano (MCU board) | DM164150 | 1.8 - 5.5 V | 1 | 350.0 | 350.0 | mA |
| | Instrumentation amplifier | INA125P | 2.7 - 36 V | 1 | 12.5 | 12.5 | mA |
| | Load cell (bridge excitation) | 3135_0 (Phidgets) | 5 V max | 1 | 5.0 | 5.0 | mA |
| | | | | | | 0.0 | mA |
| | | | | | | 0 | mA |
| | | | | | ***Subtotal*** | 367.5 | mA |
| | | | | | ***Safety Margin*** | 25% | |
| | | | | | ***Total Current Required on +5V Rail*** | 459.4 | mA |
| ***c2. Regulator or Source Choice*** | +5V linear regulator | L7805ABV-DG | 7 - 35 V | 1 | 1500 | 1500 | mA |
| | | | | | ***Total Remaining Current Available on +5V Rail*** | 1040.6 | mA |

| *-5V Power Rail (not used)* | Component Name | Part Number | Supply Voltage Range | # | Absolute Maximum Current (mA) | Total Current (mA) | Unit |
|---|---|---|---|---|---|---|---|
| | None: every part is single-supply | — | — | 0 | 0 | 0 | mA |
| | | | | | | 0 | mA |
| | | | | | | 0 | mA |
| | | | | | | 0 | mA |
| | | | | | ***Subtotal*** | 0 | mA |
| | | | | | ***Safety Margin*** | 25% | |
| | | | | | ***Total Current Required on -5V Rail*** | 0 | mA |
| ***c3. Regulator or Source Choice*** | Not used (no negative rail) | — | — | 0 | 0 | 0 | mA |
| | | | | | ***Total Remaining Current Available on -5V Rail*** | 0 | mA |

| *+3.3V Power Rail (not used)* | Component Name | Part Number | Supply Voltage Range | # | Absolute Maximum Current (mA) | Total Current (mA) | Unit |
|---|---|---|---|---|---|---|---|
| | None on this board | — | — | 0 | 0 | 0 | mA |
| | | | | | | 0 | mA |
| | | | | | | 0 | mA |
| | | | | | | 0 | mA |
| | | | | | ***Subtotal*** | 0 | mA |
| | | | | | ***Safety Margin*** | 25% | |
| | | | | | ***Total Current Required on +3.3V Rail*** | 0 | mA |
| ***c4. Regulator or Source Choice*** | Not used (no +3.3 V loads) | — | — | 0 | 0 | 0 | mA |
| | | | | | ***Total Remaining Current Available on 3.3V Rail*** | 0 | mA |

***C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining Current Available on each rail above is not negative.***

***D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails simultaneously. If you need multiple power sources, list each separately below and indicate which regulators will be connected to each supply. Confirm that the Total Remaining Current Available on each power source below is not negative.***

| *External Power Source 1* | Component Name | Part Number | Supply Voltage Range | Output Voltage | Absolute Maximum Current (mA) | Total Current (mA) | Unit |
|---|---|---|---|---|---|---|---|
| **Power Source 1 Selection** | 9 V 3 A plug-in wall adapter (switching), issued in class | BestCH 9V 3A (Amazon B09ZTKTLGW) | 100 - 240 VAC | +9V | 3000 | 3000 | mA |
| **Power Rails Connected to External Power Source 1** | +12V regulator (not used) | — | — | 0 | 0 | 0 | mA |
| | +5V linear regulator | L7805ABV-DG | 7 - 35 V | 1 | 1500 | 1500 | mA |
| | +3.3V regulator (not used) | — | — | 0 | 0 | 0 | mA |
| | | ***Total Remaining Current Available on External Power Source 1*** | | | | 1500.0 | mA |

| *External Power Source 2 (not used)* | Component Name | Part Number | Supply Voltage Range | Output Voltage | Absolute Maximum Current (mA) | Total Current (mA) | Unit |
|---|---|---|---|---|---|---|---|
| **Power Source 2 Selection** | Not used (single 9 V adapter) | — | — | — | 0 | 0 | mA |
| **Power Rails Connected to External Power Source 2** | -5V regulator (not used) | — | — | 0 | 0 | 0 | mA |
| | | ***Total Remaining Current Available on External Power Source 2*** | | | | 0 | mA |

***E. Calculate Battery Life (if applicable). For each battery, also check the worst-case lifetime of the battery by indicating the capacity in mAh.***

| | Component Name | Part Number | Supply Voltage Range | Capacity (mAh) | Required By Regulators | |
|---|---|---|---|---|---|---|
| | No battery (wall-powered board) | — | — | N/A | 0 | |
| | | | | ***Battery Life*** | N/A | |

***Notes***

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).

If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit

Absolute maximum currents are worst-case datasheet limits: PIC18F57Q43 maximum current on VDD pin 350 mA (48-pin device, -40 to +85 C); INA125P quiescent current 0.525 mA max plus 12 mA output short-circuit current; load cell 5 V excitation across its 1 kOhm bridge = 5 mA. The MCP6004 was removed: the INA125P alone sets the gain and a passive RC filter (no current draw) replaces the active filter.

Worst-case L7805ABV-DG dissipation: (9 V - 5 V) x 459 mA = 1.8 W, so the regulator gets a Boyd 507302B00000G TO-220 heatsink. Expected normal draw is under 100 mA (about 0.4 W).